

Integration by Substitution (U-Substitution)

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Name: _____

Date: _____

Learning Objectives

- Identify the appropriate substitution (u) for a composite integrand
- Rewrite an integral entirely in terms of u and du, adjusting for constant factors
- Integrate using the Power Rule and back-substitute to express the answer in terms of x

Problems

1. Use the substitution rule to find the integral of $3x^2$ squared times the square root of x^3 plus 5, with respect to x . Let u equal x^3 plus 5.

$$\int 3x^2 \sqrt{x^3 + 5} \, dx$$

2. Find the integral of $4x$ times the quantity x^2 plus 7, raised to the third power, with respect to x . Let u equal x^2 plus 7.

$$\int 4x(x^2 + 7)^3 \, dx$$

3. Use u-substitution to evaluate the integral of $5x$ to the fourth power times the quantity x^5 minus 2, raised to the fourth power, with respect to x .

$$\int 5x^4(x^5 - 2)^4 \, dx$$

4. Find the integral of x^3 times the quantity x^4 plus 1, raised to the fifth power, with respect to x . Note that you will need to adjust for a constant factor.

$$\int x^3(x^4 + 1)^5 \, dx$$

5. Use u-substitution to evaluate the integral of secant squared of $3x$, with respect to x .

$$\int \sec^2(3x) \, dx$$



6. Find the integral of secant squared of the quantity 1 over x, all divided by x squared, with respect to x. Let u equal 1 over x.

7. Evaluate the integral of 6x to the fifth power divided by the square root of x to the sixth plus 4, with respect to x.

$$\int \frac{6x^5}{\sqrt{x^6 + 4}} dx$$

8. Use u-substitution to find the integral of x squared times the quantity 2 plus x cubed, raised to the negative second power, with respect to x.

$$\int x^2(2 + x^3)^{-2} dx$$

9. Evaluate the integral of the quantity 3x squared minus 2 times the quantity x cubed minus 2x, raised to the seventh power, with respect to x.

$$\int (3x^2 - 2)(x^3 - 2x)^7 dx$$

10. Find the integral of x squared times the square root of the quantity 4 minus x cubed, raised to the fifth power, with respect to x. (Hint: rewrite the radical as a fractional exponent and adjust for constant factors.)

$$\int x^2(4 - x^3)^{5/2} dx$$



Integration by Substitution (U-Substitution) — Answer Key

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Answer Key

1. Answer: $(2/3)(x^3 + 5)^{3/2} + C$

- Let $u = x^3 + 5$, so $du = 3x^2 dx$
- Rewrite: $\int \sqrt{u} du$
- Rewrite as: $\int u^{1/2} du$
- Integrate: $u^{3/2} / (3/2) + C = (2/3)u^{3/2} + C$
- Back-substitute: $(2/3)(x^3 + 5)^{3/2} + C$

2. Answer: $(x^2 + 7)^{3/2} / 2 + C$

- Let $u = x^2 + 7$, so $du = 2x dx$, meaning $4x dx = 2 du$
- Rewrite: $\int 2u^{3/2} du$
- Integrate: $2 \cdot u^{5/2} / (5/2) + C = u^{5/2} / 2 + C$
- Back-substitute: $(x^2 + 7)^{5/2} / 2 + C$

3. Answer: $(x^5 - 2)^{3/5} / 5 + C$

- Let $u = x^5 - 2$, so $du = 5x^4 dx$
- Rewrite: $\int u^{3/5} du$
- Integrate: $u^{8/5} / (8/5) + C = u^{8/5} / 5 + C$
- Back-substitute: $(x^5 - 2)^{8/5} / 5 + C$

4. Answer: $(x^4 + 1)^{3/4} / 24 + C$

- Let $u = x^4 + 1$, so $du = 4x^3 dx$, meaning $x^3 dx = (1/4) du$
- Rewrite: $\int (1/4) u^{3/4} du$
- Integrate: $(1/4) \cdot u^{7/4} / (7/4) + C = u^{7/4} / 24 + C$
- Back-substitute: $(x^4 + 1)^{7/4} / 24 + C$

5. Answer: $(1/3) \tan(3x) + C$

- Let $u = 3x$, so $du = 3 dx$, meaning $dx = (1/3) du$
- Rewrite: $\int (1/3) \sec^2(u) du$
- Integrate: $(1/3) \tan(u) + C$
- Back-substitute: $(1/3) \tan(3x) + C$

6. Answer: $-\tan(1/x) + C$

- Let $u = 1/x = x^{-1}$, so $du = -x^{-2} dx = -(1/x^2) dx$
- Therefore $(1/x^2) dx = -du$
- Rewrite: $\int \sec^2(u) \cdot (-du) = -\int \sec^2(u) du$
- Integrate: $-\tan(u) + C$
- Back-substitute: $-\tan(1/x) + C$



**7. Answer: $2\sqrt{x^3 + 4} + C$**

- Let $u = x^3 + 4$, so $du = 3x^2 dx$
- Rewrite: $\int u^{-1/2} du$
- Integrate: $u^{1/2} / (1/2) + C = 2u^{1/2} + C$
- Back-substitute: $2\sqrt{x^3 + 4} + C$

8. Answer: $-1 / (3(2 + x^3)) + C$

- Let $u = 2 + x^3$, so $du = 3x^2 dx$, meaning $x^2 dx = (1/3) du$
- Rewrite: $\int (1/3) u^{-2} du$
- Integrate: $(1/3) \cdot u^{-1} / (-1) + C = -1/(3u) + C$
- Back-substitute: $-1 / (3(2 + x^3)) + C$

9. Answer: $(x^3 - 2x)^{1/8} / 8 + C$

- Let $u = x^3 - 2x$, so $du = (3x^2 - 2) dx$
- Rewrite: $\int u^{1/8} du$
- Integrate: $u^{9/8} / (9/8) + C$
- Back-substitute: $(x^3 - 2x)^{1/8} / 8 + C$

10. Answer: $-2(4 - x^3)^{7/2} / 21 + C$

- Let $u = 4 - x^3$, so $du = -3x^2 dx$, meaning $x^2 dx = -(1/3) du$
- Rewrite: $\int u^{5/2} \cdot (-1/3) du = -(1/3) \int u^{5/2} du$
- Integrate: $-(1/3) \cdot u^{7/2} / (7/2) + C = -(1/3) \cdot (2/7) u^{7/2} + C$
- Simplify: $-2u^{7/2} / 21 + C$
- Back-substitute: $-2(4 - x^3)^{7/2} / 21 + C$

